LEARNING OPTIMIZATION · BHT BERLIN

Dynamic Pricing Competition

Presentation #1 — Linear regression & capacity utilization curves

Simon & Alois · 3 November 2026

AGENDA

Where we are after four weeks

1. Goals and current status
2. Capacity utilization curves
3. Demand model by linear regression
4. Performance evaluation
5. What worked, what didn’t, next steps

What we want the algorithm to do

Beat the competitor

The ranking counts, not the revenue alone: more revenue than the other airline in each simulation.

Sell 80 seats at the right pace

No stockout long before day 100, no empty seats left at departure.

Respect the platform

Answer in 0.2 s on average, no file writes, 99% of requests error-free.

Status: one shared structure, two ways to learn the demand. **Simon** learns it offline, from past competitions. **Alois** re-learns it online, during each season. Both are on the DPC.

Our pricing algorithm in three steps

1

Target curve

How many seats should be sold by today?

2

Demand model

Linear regression gives the revenue-maximizing price p*.

3

Correction

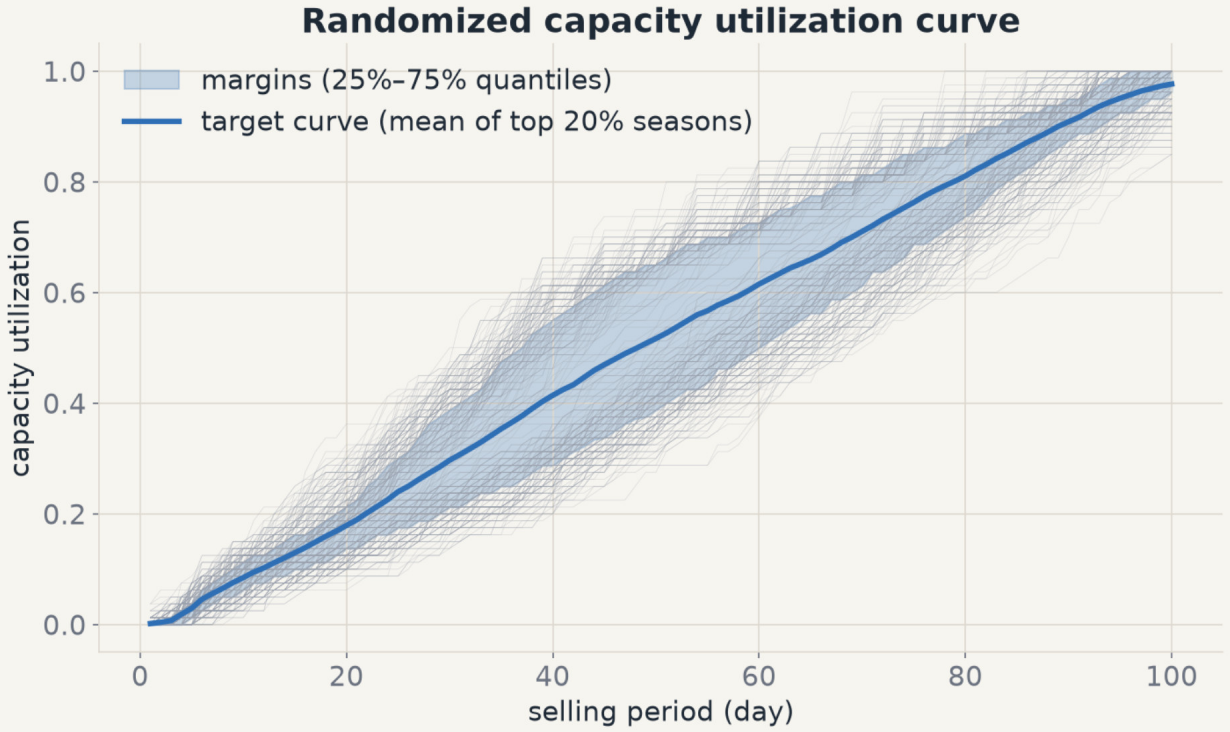
Too fast → raise the price. Too slow → lower it.

Special cases: competitor sold out (monopoly rule), our own stockout, safety nets when seats run out too fast.

Same start, two directions

Week	Simon — learn offline	Alois — learn online
W2	Shared base: follow a target curve, reference price = 50% last price + 50% competitor price	
W3	OLS fitted on one past competition, optimal price p*	OLS re-fitted every update on the current season, 50% p* + 50% curve price
W4	OLS pooled on 16 competitions, target margins, ARIMA, model-based monopoly price	ARX(1) with yesterday's demand, 40% p*, weaker competitor anchor, scarcity monopoly price

A target curve with margins

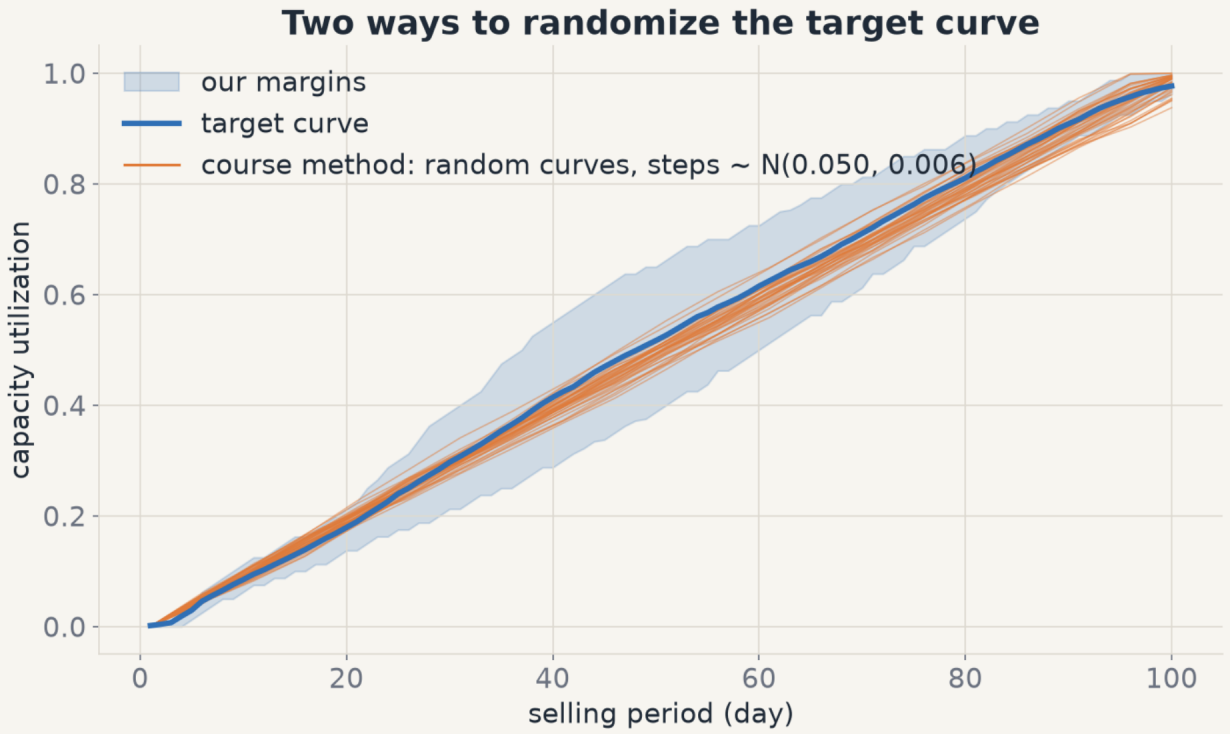


Target = mean cumulative sales of the 20% best-paying seasons (16 competitions).

Margins = 25% / 75% quantiles. Day 50: **31 · 41 · 52 seats**.

Sales are Poisson-random: inside the margins we **keep the price**.

Randomized curves: course method vs. margins



Course method: 5-day steps drawn from N(0.050, 0.006), a new curve every season.

Random steps cancel out: only **±1.4 seats** at day 50.

Simulated revenue: 2080 vs. **2096 with margins**. We keep the margins; the course method stays an option.

From a regression to an optimal price

$$\text{demand} = a + b \cdot \text{price} + c \cdot \text{competitor price}$$

Ordinary least squares · $b < 0$ (our price), $c > 0$ (competitor price)

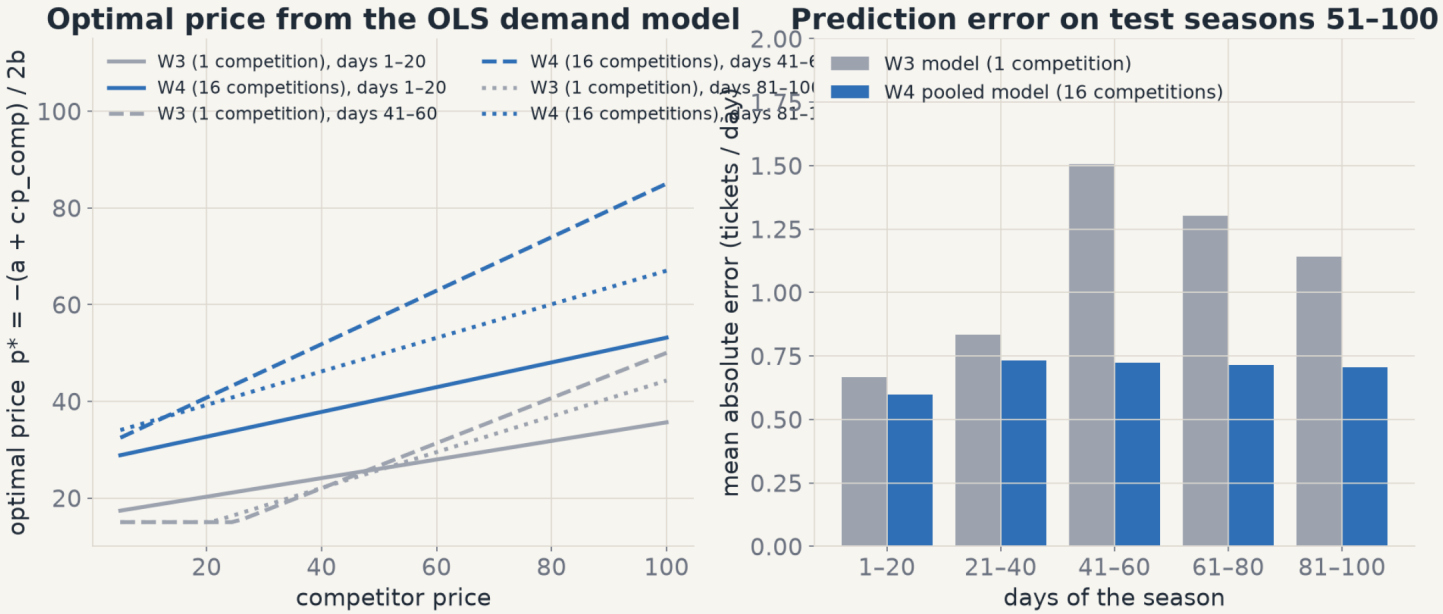
Revenue is a parabola

$R(p) = p \cdot (a + c \cdot p_comp + b \cdot p)$, maximum at **$p^* = -(a + c \cdot p_comp) / 2b$**: exact and instant, used by both of us.

The open question

Which data estimate a, b, c? **Past competitions** (offline) or **the current season** (online)?

Offline: one competition vs. 16 pooled



W3, one competition: p* ≈ 16–25 € — trained on a cheap market, too cheap elsewhere.

W4, 16 competitions: mean absolute error 0.60–0.73 vs. 0.67–1.51 tickets/day on seasons 51–100; p* ≈ 33–70 €.

R^2 only 0.01–0.07: daily demand is mostly Poisson noise.

Online: re-estimate during the season

$$demand_t = b0 + b1 \cdot p_{own} + b2 \cdot p_{comp} + \varphi \cdot demand_{t-1}$$

ARX(1), inspired by ARIMA · least squares on the current season's history, from day 7

Price

$p^* = -(b0 + b2 \cdot p_{comp} + \varphi \cdot d_{t-1}) / 2 \cdot b1$ if $b1 < -0.01$, then **40% p^* + 60% target-curve price**.

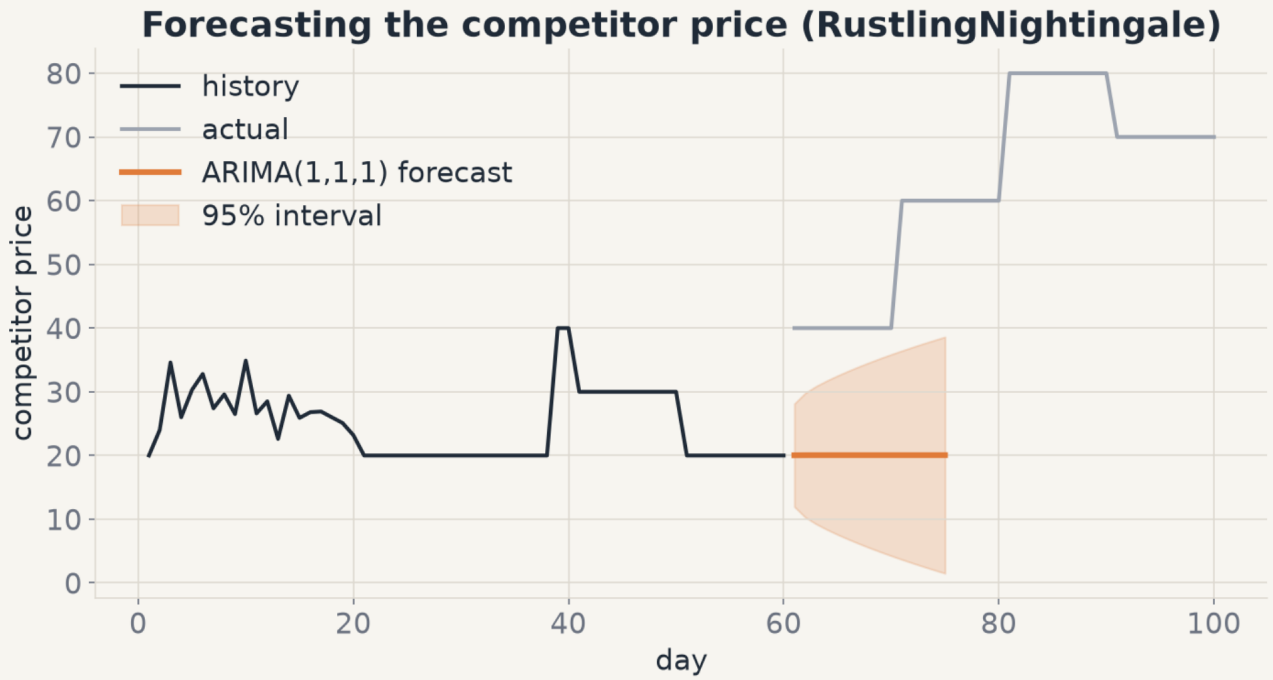
+ Adapts

Learns the demand of **this** simulation, which is random on the DPC.

− Little data

At most 100 points per season, mostly 0 or 1 ticket, and our price must vary.

Which competitor price goes into p*?

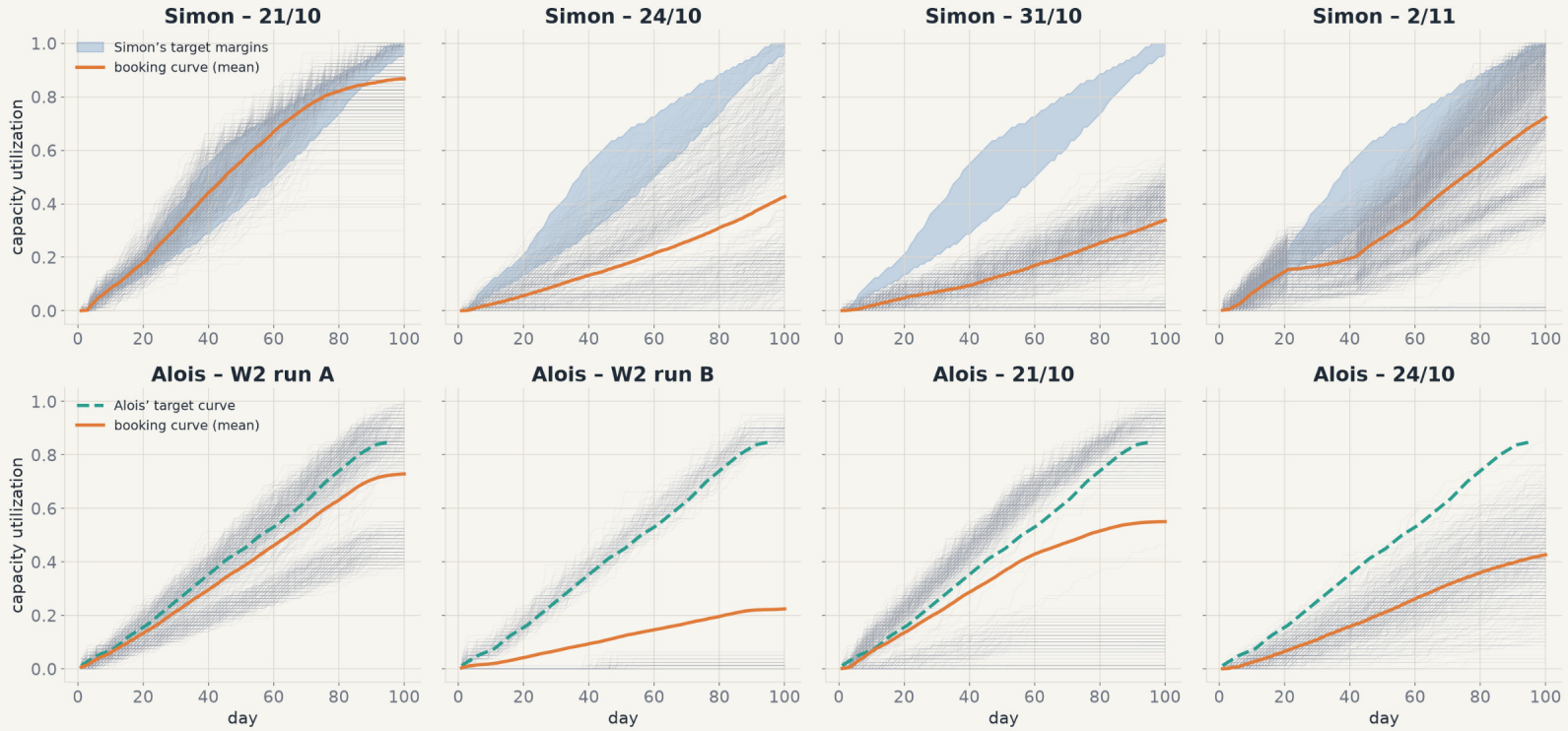


ARIMA(1,1,1) on the competitor price, within the season (18 ms on average, 74 ms max per fit).

Error **2.50 €** vs. **2.47 €** for the last observed price.

Prices move in steps: the last observed price is already a strong forecast.

Target curves vs. booking curves on the DPC

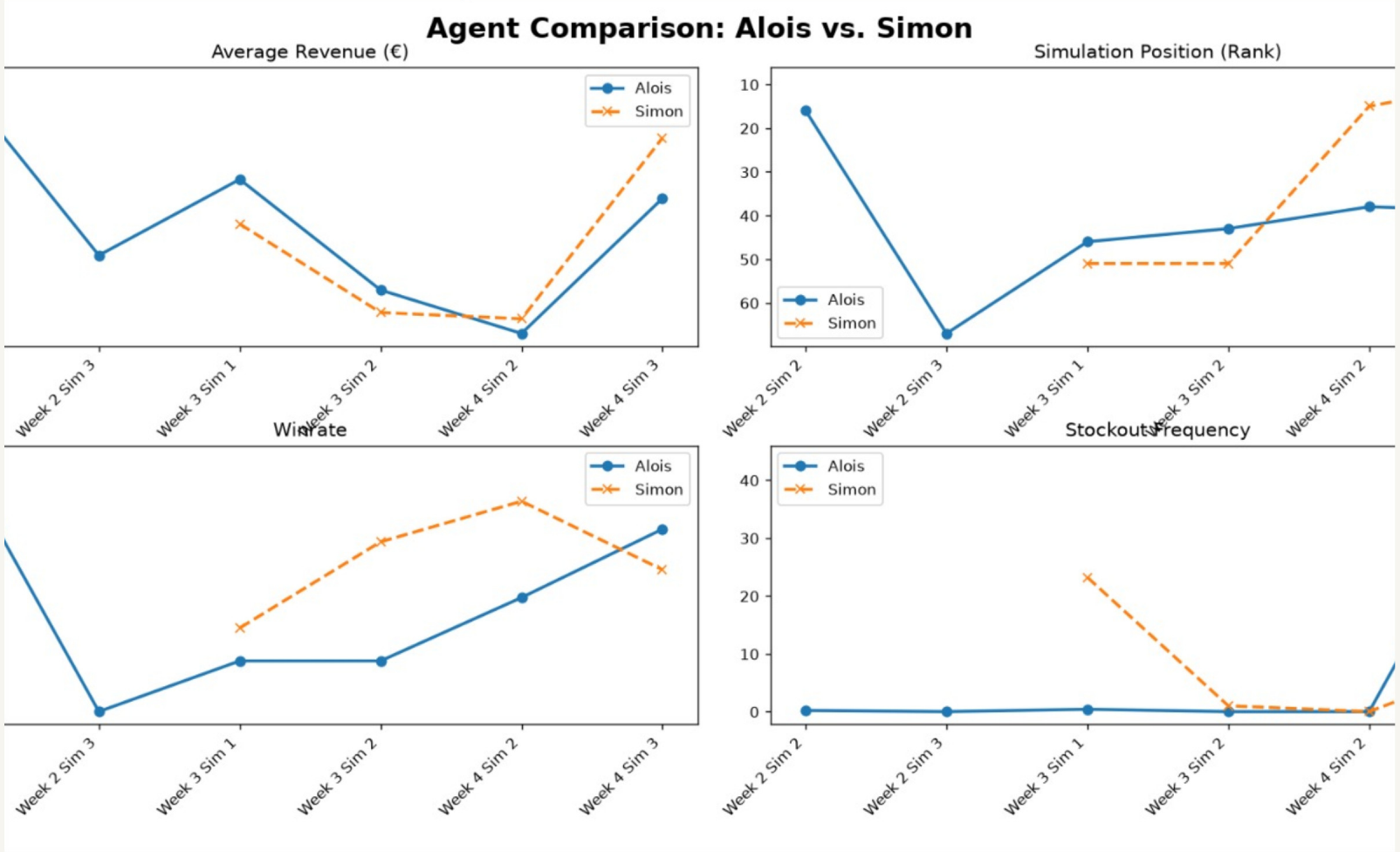


21/10, both of us (W3): on track until the competitor sells out, then flat: 0–3 tickets in monopoly at ~130 €.

Low-demand runs (24/10, 31/10, Alois W2 B): the target is out of reach and prices drift down.

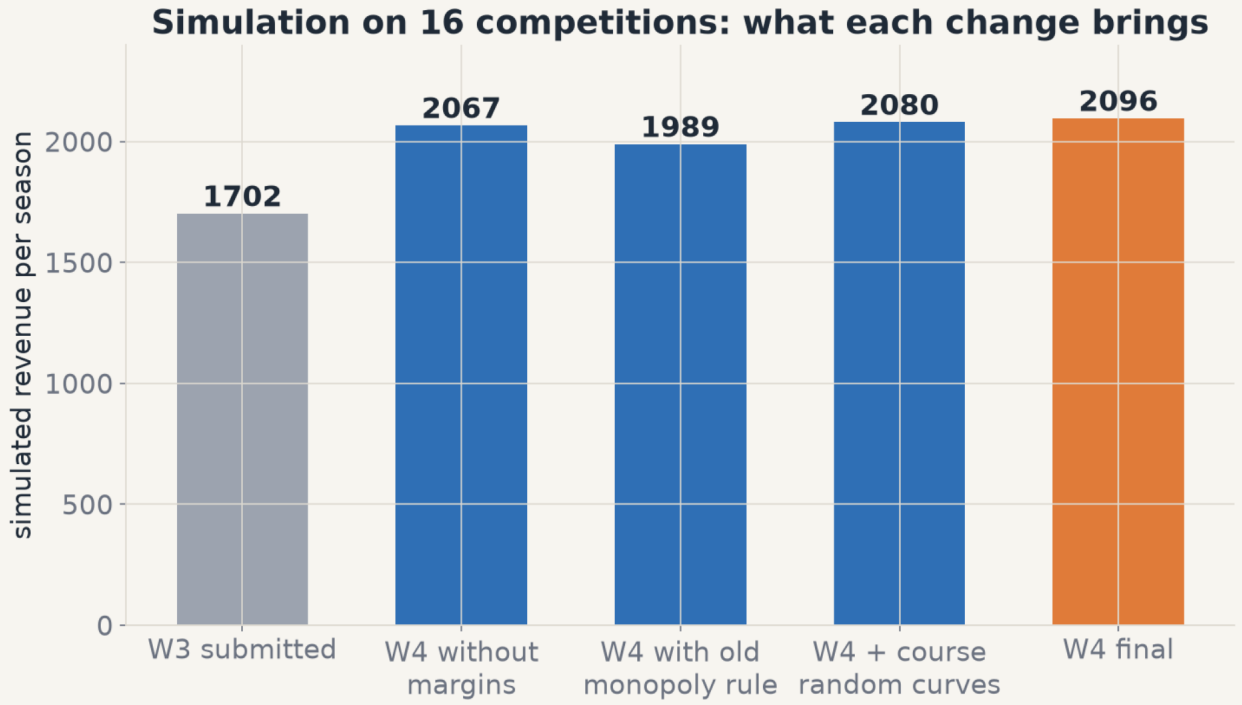
2/11, Simon W4: new monopoly rule sells 2009 tickets at ~51 €. **Lesson:** the target must adapt to demand.

Results on the DPC platform



Each of us faced different competitors and demands: **these numbers cannot rank our algorithms** — hence a common simulator.

Our simulator: what each W4 change brings

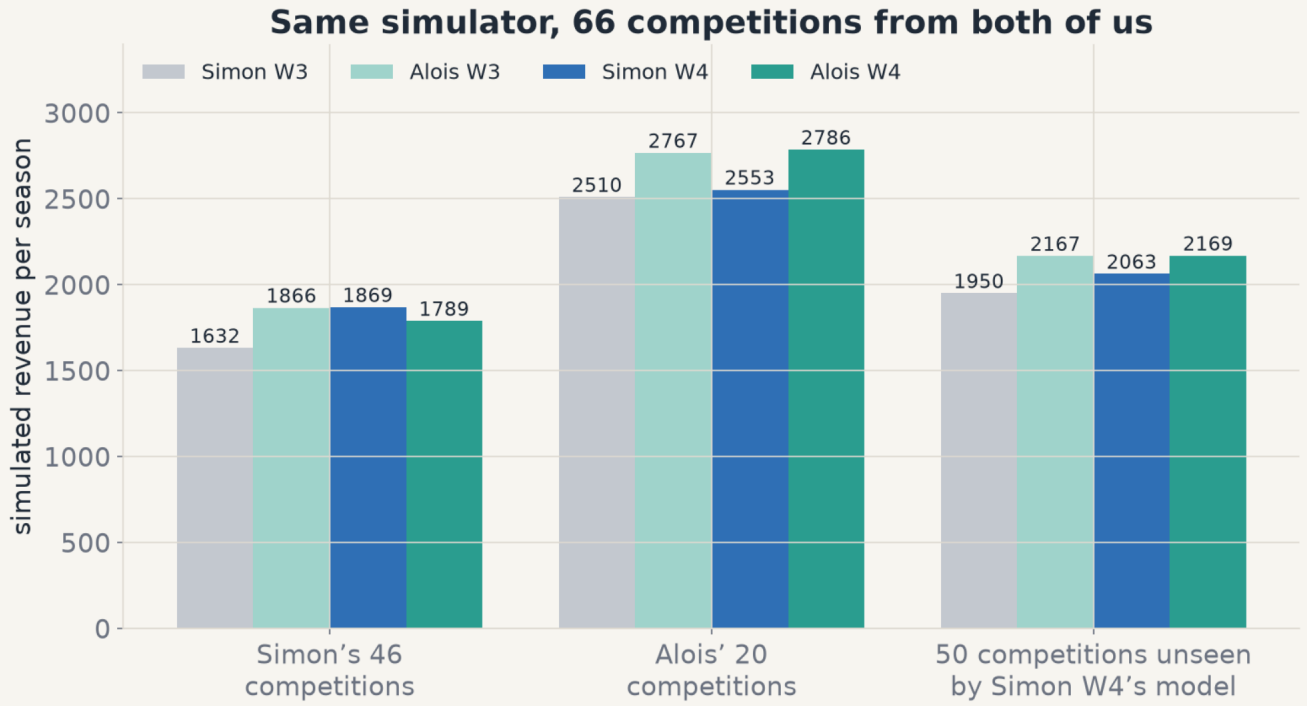


Setup: one OLS demand model per competition (16 “worlds”), competitor prices replayed, Poisson demand, 80 seats.

Simon W4: +23% vs. W3 on his 16 competitions — pooled OLS and monopoly rule; margins +1.4%.

Limit: the simulated competitor never reacts to our price.

Offline vs. online: a fair comparison



Home advantage: each algorithm wins on its author's data.

Unseen data: Alois W4 **2169** vs. Simon W4 **2063** (+5%, 57% of seasons won).

W3 → W4 on unseen data: Simon +6%, Alois stable (2167 → 2169).

Ideas that worked, ideas that didn't

Worked

- Closed-form optimal price p*
- Online regression: best on unseen data
- Pooled regression: better than 1 competition
- Margins around the target (+1.4%)
- Model-based monopoly price (2/11: 2009 tickets at ~51 €)
- Our own simulator to compare fairly

Didn't work

- Monopoly at ~130 € (W3): 0–3 tickets, both of us
- Fixed target in low-demand runs
- Offline model from one competition
- ARIMA: no better than the last observed price
- Course random curves: spread too small

5 · NEXT STEPS

Where we go next

Problem	Next step	Course
80 seats ignored in p*	Capacity-constrained price optimization	10 Nov
Offline vs. online model	Pooled model as a start, updated online	17 Nov
Competitor sold out	Exploit the monopoly phase properly	15 Dec
Fixed target in low demand	Scale the target to the observed demand level	5 Jan

DPC · PRESENTATION #1

Thank you — questions?

2 ways

to learn the demand: offline and online

66

competitions in a common simulator

1 plan

combine both approaches

Simon & Alois